

T H E D A T A F L O W C O M P A N Y

Consumer Decision Making Analysis Using R

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Agenda

- **Day 1: Foundations**

- Introduction to the reproducible building blocks of a open-science eco-system: GitHub, README files, decision logs, and data protection principles
- Creating a clear folder structure for data, scripts, outputs, reports, and documentation
- Setting up a reproducible RStudio Project
- Introduction to R and RStudio
- Basic R concepts: scripts, console, objects, functions, packages, and dataframes

→ After Day 1, you will know which software you can use and how to set up a fully integrated and reproducible project repository.

- **Day 2: Applied data analysis**

- How do we code and learn these days
- Importing, inspecting, cleaning, and transforming data
- Recoding variables and handling missing values
- Creating descriptive tables and visualisations
- Running group comparisons and regression models
- Introduction to moderation and mediation analysis
- Practising a simple collaborative GitHub workflow

→ After Day 2, you will know how to fill up the project repository with R code and feel familiar using it.

- **Day 3: Reporting and reproducible outputs/ The power of LLMs**

- Exporting tables, figures, and model results
- Writing short methods and results sections from code
- Introduction to Quarto reports
- Integrating figures and tables into a compact report
- Reviewing reproducibility, data protection, and responsible AI use

→ After Day 3, you will know how to create shiny reports (and outputs in general) and how to integrate newest LLM approaches into your reproducible project repository (Day 1) and codebase (Day 2).

- **Participant requirements**

- Participants should bring a laptop
- R and RStudio should ideally be installed beforehand, although installation can also be supported during the workshop
- A GitHub account is recommended but not strictly required
- Participants can bring their own data; the workshop will use a simulated teaching dataset

- **Expected outcome**

- By the end of the workshop, participants should have a structured R project, cleaned data, basic analyses, visualisations, exported outputs, and a compact report template they can adapt for future research projects.